

Individual Assignment 3
Due at the beginning of the class on Feb. 25
Total: 5 points

East-West Airlines has entered into a partnership with the wireless phone company Telcon to sell the latter's service via direct mail. "phone_sale.csv" is a sample of data, provided so that the analyst can develop a model to classify East-West customers as to whether they purchase a wireless phone service contract.

In this problem, we use **Logistic Regression** to classify the **target** variable (**Phone_sale**). The variable description is as follows:

Field Name	Description
Topflight	Indicates whether flyer has attained elite "Topflight" status, 1 = yes, 0 = no
Balance	Number of miles eligible for award travel
Qual_miles	Number of miles counted as qualifying for Topflight status
Bonus_miles	Number of miles earned from non-flight bonus transactions in the past 12 months
Bonus_trans	Number of non-flight bonus transactions in the past 12 months
Flight_miles_12mo	Number of flight miles in the past 12 months
Flight_trans_12	Number of flight transactions in the past 12 months
Club_member	Member of the airline's club (paid membership), 1=yes, 0=no
Any_cc_miles_12mo	Variable indicating whether member added miles on any credit card type within the past 12 months (1='Y', 0='N')
Phone_sale	Target variable indicating whether member purchased Telcom service as a result of the direct mail campaign

1. Preprocess the data. Re-code the target value “**Phone_sale = Yes**” to be class
1. Transform **Any_cc_miles_12mo** to a categorical variable with levels renamed as ‘Y’ and ‘N’, and set the base level to ‘Y’.
2. We choose to use only **Bonus_trans** and **Any_cc_miles_12mo** as predictors and drop other variables.
3. Partition the data using the pre-generated training indexes provided in “a3_TrainIndex.rda” (generated with the seed 111 for 60% training vs. 40% validation). Run logistic regression on training data.
4. Find the fitted coefficient values. Write down the model:

logit =

5. Interpret the result. Based on the fitted coefficient values, what can you say about the effect of “**Any_cc_miles_12mo**”?
6. Use the above logistic regression model to calculate the probability for (**Phone_sale = Yes**) if:

Bonus_trans = 50, Any_cc_miles_12mo = ‘Y’

Prob(Phone_sale = Yes) = _____

(HINT: you can either use R codes to find the predicted probability, OR compute the probability manually using the fitted coefficients.)

If the cutoff value is set to 0.3, is the above customer predicted to purchase a wireless phone service contract?

7. Set the cutoff value to be 0.5. Write down the confusion matrix of the **validation** data.

8. Our goal is to identify potential customers that will purchase a wireless phone as a result of the direct mail campaign. Based on the confusion matrix you got in step 7, is our current logistic regression model useful for this task? Why?

9. To deal with the above issue, should the cutoff value be increased or decreased? Why?